

CYBERNEXUS X

Demo Cheat Sheet v2 — Sample Inputs for a Live Walkthrough

This sheet walks one incident end-to-end through every module — detect → assess → enrich → analyze → respond → preserve → report — with copy-pasteable example values for each screen. All IPs below use RFC 5737 documentation ranges (192.0.2.0/24, 198.51.100.0/24, 203.0.113.0/24) — safe, non-routable addresses meant exactly for demos like this.

Login — Two Roles, Two Logins

CYBERNEXUS X has exactly two roles. The website's own **Register** tab always creates an **analyst** — there is no role picker in the UI, by design. Compliance, deleting assets, and manually blocking an IP are **admin-only**, enforced both in the UI and on the server (a non-admin request is rejected with HTTP 403 even if it bypasses the UI).

Role	How to get it	Can do
analyst	Register tab in the UI	Dashboard, Scanning, Threats, AI Analyst, Incidents, Evidence
admin	API call with role:"admin" (below)	Everything analyst has, + Compliance, Delete asset, Block IP manually

To log in as admin — run this once against your live URL (swap the domain, email, password, name). One line, no backtick continuations — safe to paste directly into PowerShell:

```
Invoke-RestMethod -Method Post -Uri "https://cybernexus-x.onrender.com/api/auth/register"
-ContentType "application/json" -Body
'{"email":"you@example.com", "password":"YourStrongPassword123", "name":"Admin", "role":"admin"}'
```

A successful response includes "role":"admin". Then sign in through the normal login screen with that email/password — the Compliance link appears in the sidebar immediately.

Optional — Enable the API-Powered Features

Every AI / threat-intel feature is **real, not mocked** — and degrades honestly to “unavailable” (never fabricates) if a key below is unset or an Anthropic account has no credit. Set these once in **Render** → **your service** → **Environment**, then redeploy.

Env var	Powers	Get a free key at
ANTHROPIC_API_KEY	AI Threat Analyst	console.anthropic.com — needs billing/credit added
ABUSEIPDB_API_KEY	Threat intel lookup	abuseipdb.com/account/api — free, 1,000/day
VIRUSTOTAL_API_KEY	Threat intel lookup	virustotal.com profile → API Key — free, 500/day

The Workflow

STEP 1 · DASHBOARD

Detect — Ingest logs, watch threats stream in live

Click **“Ingest sample logs.”** Built-in sample data — no input needed. Threat events (syn-flood, brute-force, port-scan, anomaly-rate) appear over the live Socket.io feed and populate every chart on the page.

“Nothing here is seeded — every number traces to a real database record. Watch it fill up as we run an actual detection pipeline.”

STEP 2 · SCANNING

Assess — Add an asset, scan it, delete it (admin)

Add an asset, then click **Scan**. Admins also see a **Delete asset** button per row — it asks for confirmation before removing the asset and its scan/vulnerability history.

```
Add Asset
name Local Web Server
type server
ipAddress 127.0.0.1
authorizedForScan true
```

```
Run Scan
startPort 1
endPort 1024
```

“Scanning never leaves localhost or the authorized ranges. The CVSS numbers come from the real FIRST.org v3.1 formula — verified against the official test vectors — not a lookup table.”

STEP 3 · THREATS

Enrich — Look up a source IP in threat intelligence

Use the intel lookup box with a documentation-range IP (safe to query against AbuseIPDB / VirusTotal caches).

```
Indicator to look up: 198.51.100.23
```

“Real free-tier threat-intel APIs, cached in Mongo. No API key configured? It reports ‘unknown’ instead of fabricating a verdict.”

STEP 4 · AI ANALYST

Analyze — Ask the AI Threat Analyst for an assessment

Click **“Run AI analysis.”** No input needed — Claude reasons over the threats already detected in Step 1 and returns severity, an attack narrative, correlated IPs, and prioritized remediation steps.

If you see: “the connected Anthropic account has no API credit” — add billing at console.anthropic.com. This confirms the whole pipeline works; it's an account-balance issue, not a bug.

“The AI is grounded strictly in real detected data — it can't invent threats. If the key is unset it shows ‘unavailable’; if the Anthropic account has no credit, it now shows a clear billing message instead of raw API JSON — never a fabricated result.”

STEP 5 · INCIDENTS

Respond — Auto-correlate an incident, then contain it

Click **“Run auto-response.”** Unacknowledged high-severity threats sharing a source IP fold into one incident and get auto-blocked (simulated). Admins can also block an IP manually. Walk the incident through its status workflow.

```
Auto-response
autoBlock true
minScore 70 (default)
```

```
Manual block (admin only)
ip 203.0.113.55
reason Manual block from console
```

```
Status walk: new → investigating → contained → resolved
```

“Containment is gated behind an explicit live-mode toggle, so nothing disruptive runs by accident — the real firewall stays off unless you flip it on.”

STEP 6 · EVIDENCE

Preserve — Append tamper-evident evidence

Add a couple of records. Watch the chain-integrity banner stay ● **CHAIN VERIFIED** as the ledger grows.

```
Record 1
type note
payload { "note": "collected during triage", "analyst": "SOC Admin" }
```

```
Record 2
type scan
payload { "assetIp": "127.0.0.1", "findings": 3 }
```

“This is a real SHA-256 hash-chain ledger — each record’s hash feeds the next. Tamper with any past payload and verification breaks for that record and everything after it.”

STEP 7 · COMPLIANCE (ADMIN ONLY)

Report — Generate the compliance PDF

Review the audit trail, then click **“Download PDF report.”** No input needed. This page and its API are invisible/blocked to non-admins by design.

“Every figure in this PDF is pulled from immutable audit records and the hash-chained ledger — an auditor-grade artifact, generated on demand.”

Quick Reference

Page	Route	Key action	Role
Dashboard	/	Ingest sample logs	any
Scanning	/scanning	Add asset → Scan → Delete	analyst / admin*
Threats	/threats	Intel lookup on an IP	any

Page	Route	Key action	Role
AI Analyst	/ai	Run AI analysis	any
Incidents	/incidents	Auto-respond → Block IP	analyst / admin*
Evidence	/evidence	Append record → Re-verify chain	any
Compliance	/compliance	Download PDF	admin only

* base action available to any role; Delete asset / Block IP require admin.

Safe Demo IP Ranges (RFC 5737)

Range	Example used above	Purpose
192.0.2.0/24	192.0.2.10	TEST-NET-1
198.51.100.0/24	198.51.100.23	TEST-NET-2 (intel lookup)
203.0.113.0/24	203.0.113.55	TEST-NET-3 (manual block)

What to Emphasize

- **No fabricated data** — every dashboard number is a real DB record; AI and threat-intel degrade honestly when API keys or credit are absent, never inventing a result.
- **Real crypto / security primitives** — genuine CVSS v3.1 math and a genuine tamper-evident SHA-256 hash chain, not simulations.
- **Defense in depth on roles** — admin-only actions (Compliance, Delete asset, Block IP) are gated in the UI *and* re-checked on the server, so nothing depends on hiding a button.
- **Clean failure states** — a missing AI key, an empty Anthropic credit balance, or a missing threat-intel key all produce clear, actionable messages instead of raw errors or fake data.
- **Safe by default** — scanning is localhost/private-range only; IP blocking is simulated behind a manual live-mode gate.
- **One coherent vertical slice** — a single attack flows cleanly from detection all the way to a signed compliance report.

CYBERNEXUS X — Security Operations Center platform · v2 cheat sheet · Reset between demos: the in-memory dev DB wipes on restart; on the deployed (Atlas) instance, just re-ingest sample logs.